

particularly invertebrates—as they grow from juvenile to adult. In insects, metamorphosis can be complete or incomplete. Complete metamorphosis involves a total reorganization of the organism during a resting stage, called a pupa. Incomplete metamorphosis embraces a series of less drastic changes; these occur each time the young animal molts. See also *Larva*, *Nymph*.

METATARSAL In 4-limbed animals, one of a set of bones in the hindleg, forming a joint with a digit distally. In most primates, the metacarpals form the sole of the foot.

METATHORAX The second segment of an insect's thorax.

MIGRATION A journey to a different region, following a well-defined route. Most animals that migrate do so in step with the seasons, so they can take advantage of good breeding conditions in one place, and good wintering ones in another. See also *Partial migrant*.

MIMICRY A form of camouflage in which an animal resembles another animal or an inanimate object, such as a twig or a leaf. Mimicry is very common in insects, with many harmless species imitating ones that have dangerous bites or stings.

MOLAR TOOTH In mammals, a tooth at the rear of the jaw. Molar teeth may have a flattened or ridged surface for chewing vegetation. The sharper molar teeth of meat-eaters cut through hide and bone.

MOLT Shedding fur, feathers, or skin so that it can be replaced. Mammals and birds molt to keep their fur and feathers in good condition, to adjust their insulation, or so that they can be ready to breed. Arthropods—such as insects—molt in order to grow.

MONOGAMOUS Mating with a single partner, either during the course of one breeding season or throughout life. Monogamous partnerships are common in animals that care for their young.

MORPH A physical variant of a species, sometimes known as a phase. Some species have several clearly defined morphs, which may differ in color or patterning. See also *Polymorphic*.

MUTUALISM Two organisms that interact to the mutual benefit of both.

NEUROTOXIN A substance that deactivates nerves or that disrupts the way that they work. Neurotoxins can be lethal in very small amounts. They are found in snake venom and in poisons produced by some other animals, such as frogs, toads, fishes, and insects.

NEW WORLD The Americas.

NICHE An animal's place and role in its habitat. Although 2 species may share the same habitat, they never share the same niche.

NICTITATING MEMBRANE In birds and many reptiles—apart from snakes—a transparent or semiopaque “third eyelid,” which moves sideways across the eye. Aquatic birds often use the membrane as an aid to vision when swimming underwater.

NOCTURNAL Active at night.

NOTOCHORD A reinforcing rod that runs the length of the body. The notochord is a distinctive feature of chordates, although in some it is present only in early life. In vertebrates, the notochord becomes incorporated in the backbone during the development of the embryo.

NUCHAL CREST A crest at the nape of the neck.

NYMPH An immature insect that looks similar to its parents but that does not have functioning wings or reproductive organs. A nymph develops the adult shape by metamorphosis, changing slightly each time it molts.

OCELLUS (pl. OCELLI) In insects, a small simple eye that detects overall levels of light. Many insects have 3 ocelli, positioned in a cluster on top of the head.

OLD WORLD Europe, Asia, Africa, and Australasia.

OMNIVORE An animal that eats both plant and animal food.

OOTHECA (pl. OOTHECAE) An insect egg case. In some insects—for example, cockroaches—the ootheca has a thin covering, but in others, it is surrounded by a jacket of foam that hardens soon after it has been produced.

OPERCULUM A cover or lid. In some gastropod mollusks, an operculum is used to seal the shell when the animal has withdrawn inside. In bony fishes, an operculum on each side of the body protects the chamber containing the gills.

ORDER A level used in classification. In the sequence of classification levels, an order forms part of a class, and is subdivided into one or more families.

ORGAN A structure in the body, composed of several kinds of tissues, that carries out specific tasks.

OSSICLE A minute bone. The ear ossicles of mammals, which transmit sound from the eardrum to the inner ear, are the smallest bones in the body.

OVIPAROUS Reproducing by laying eggs. See also *Ovoviviparous*, *Viviparous*.

OVIPOSITOR In female insects, an organ at the end of the abdomen, used for laying eggs. The ovipositor often has a tubular or bladelike shape, so that it can insert the eggs in the ground, in plants, or in other animals. In some insects, the ovipositor is modified to form a sting.

OVOVIVIPAROUS Reproducing with eggs that hatch within the female's body so the young one is born alive. The eggs contain all the nutrients that the embryos need, and there is no direct connection between them and the mother. See also *Oviparous*, *Viviparous*.

PAPILLA (pl. PAPILLAE) A small, fleshy protruberance on an animal's body. Papillae often have a sensory function—for example, detecting chemicals that help pinpoint food.

PARAPODIUM (pl. PARAPODIA) A leg- or paddlelike flap found in some worms.

Parapodia are used for moving, or for pumping water past the body.

PARASITE An animal that lives on or inside another animal, and that feeds either on its host animal or on food that its host has swallowed. The majority of parasites are much smaller than their hosts, and many have complex life cycles involving the production of huge numbers of eggs. Parasites often weaken their hosts, but generally do not kill them. See also *Ectoparasite*, *Endoparasite*, *Parasitoid*.

PARASITOID An animal that is a parasite during its early life but that becomes free-living as an adult. Parasitoids are common among insects. Unlike true parasites, parasitoids usually kill their hosts—typically by eating them.

PARATOID GLAND In amphibians, a gland behind the eyes that secretes poison onto the surface of the skin.

PARTHENOGENESIS A form of reproduction in which an egg cell develops into a young animal without having to be fertilized. Parthenogenesis is common in invertebrates, and it produces offspring that are genetically identical to the parent. In animals that have separate sexes, young produced by parthenogenesis are always female.

PARTIAL MIGRANT A species in which some populations migrate, while others are sedentary. This situation is common in birds with a wide distribution, because the different populations often experience very different climatic conditions. See also *Migration*.

PATAGIUM In bats, the flap of double-sided skin that forms the wing. This term is also used for the parachute-like skin flaps of colugos and other gliding mammals.

PECTORAL FIN One of the 2 paired fins positioned toward the front of a fishes's body, often just behind its head. Pectoral fins are usually highly mobile and are normally used for maneuvering.

PECTORAL GIRDLE In 4-limbed vertebrates, the collection of bones that anchors the front limbs to the backbone. In most mammals, the pectoral girdle consists of 2 clavicles, or collarbones, and 2 scapulae, or shoulder blades.

PEDIPALPS In arachnids, the second pair of appendages, near the front of the body. Depending on the species, they are used for walking, sperm transfer, or attacking prey.

PELVIC FINS The rear paired fins in fishes, which are normally positioned close to the underside, sometimes near the head but more often toward the tail. Pelvic fins are generally used as stabilizers.

PELVIC GIRDLE In 4-limbed vertebrates, the collection of bones that anchors the front limbs to the backbone. The bones of the pelvic girdle are often fused, forming a weight-bearing ring called the pelvis.

PELVIC SPURS Spurs near the rear of the body that some male sharks and rays use during mating. Also known as claspers.

PHASE see *Morph*.

PHEROMONE A chemical produced by one animal that has an effect on other members of its species. Pheromones are often volatile substances that spread through the air, triggering a response from animals some distance away.

PHOTOSYNTHESIS A collection of chemical processes that enable plants to capture the energy in sunlight and convert it into chemical form.

PHYLUM (pl. PHYLA) A level used in classification. In the sequence of classification levels, a phylum forms part of a kingdom, and is subdivided into one or more classes.

PINNA (pl. PINNAE) The external ear flaps found in mammals.

PLACENTA An organ developed by an embryo animal that allows it to absorb nutrients and oxygen from its mother's bloodstream, before it is born. Young that grow in this way are known as placental mammals.

PLACENTAL MAMMAL see *Placenta*.

PLANKTON Floating organisms—many of them microscopic—that drift in open water, particularly near the surface of the sea. Planktonic organisms can often move, but most are too small to make any headway against strong currents. Planktonic animals are collectively known as zooplankton.

PLANTIGRADE A gait in which the sole of the foot is in contact with the ground. See also *Digitigrade*, *Unguligrade*.

PLASTRON The lower part of the shell in tortoises and turtles.

POLYANDROUS Relating to a reproductive system, rare in the animal kingdom, in which females mate with several males during the course of a single breeding season, and the male takes care of the young; seen only in some fishes and some shorebirds.

POLYGAMOUS Relating to a reproductive system in which individuals mate with more than one partner during the course of a single breeding season.

POLYGYNOUS Relating to a reproductive system in which males often mate with several females during the course of a single breeding season.

POLYMORPHIC Relating to a species that occurs in several different forms, known as morphs. The morphs may be geographically separated, so that there is only one in any particular place, or they may occur together, often in set proportions. Morphs persist even though parents from different morphs often interbreed.

POLYP In cnidarians, a body form that has a hollow cylindrical trunk, ending in a central mouth surrounded by a circle of tentacles. Polyps are frequently attached to solid objects by their base. See also *Medusa*.

PREDATOR An animal that catches and kills others, known as its prey. Some predators catch their prey by lying in wait, but most actively pursue and attack other animals.

PREHENSILE Able to curl around objects and grip them.

PREMOLAR TOOTH In mammals, a tooth positioned midway along the jaw, between the canines and the molars.

PREPUPA (pl. PREPUPAE) A resting stage that some insects pass through before forming a pupa.

PREY Any animal that is eaten by a predator.

PROBOSCIS An animal's nose, or a set of